A5

Q1

Define the state as the number of broken machines, so the state space is $\{0, 1, 2, 3\}$:

- **State 0**: All three machines are working.
- **State 1**: One machine is broken.
- **State 2**: Two machines are broken.
- **State 3**: All three machines are broken.

Transition Rates

- **State 0**: All machines are operational. The time until the next breakdown is the minimum of three independent exponential times with rates μ_1, μ_2, μ_3, so the total breakdown rate is $\mu_1 + \mu_2 + \mu_3$, transitioning to state 1. No repairs occur (repair rate = 0).
- **State 1**: One machine is broken, and the repairman is fixing it at rate μ, transitioning back to state 0. Meanwhile, two machines are working, but the breakdown rate depends on *which* machine is broken:
  - If machine 1 is broken, machines 2 and 3 can break down, rate = $\mu_2 + \mu_3$.
  - If machine 2 is broken, rate = $\mu_1 + \mu_3$.
  - If machine 3 is broken, rate = $\mu_1 + \mu_2$.
- **State 2**: Two machines are broken; the repairman fixes one at rate μ, transitioning to state 1, while the one working machine can break down.
- **State 3**: All machines are broken; the repair rate is μ to state 2, and the breakdown rate is 0.

- **Death rates**: When $n \geq 1$, the repairman works at rate μ, so $\mu_n = \mu$ for $n = 1, 2, 3$, and $\mu_0 = 0$. This is consistent.
- **Birth rates**:
  - $\lambda_0 = \mu_1 + \mu_2 + \mu_3$.
  - λ_1: Depends on which machine is broken.
  - λ_2: Depends on which two are broken.

Since λ_1 and λ_2 are not uniquely determined by the state n due to differing μ_i, the process violates the birth and death process requirement. The transition rates depend on the specific configuration of broken machines, not merely their count.

Special Case: Identical Machines

If $\mu_1 = \mu_2 = \mu_3 = \lambda$, the breakdown rates become uniform. Define state n as the number of broken machines:

- **Birth rate**: $\lambda_n = \lambda \times (3 - n)$, since $3 - n$ machines are working.
- **Death rate**: $\mu_n = \mu$ for $n \geq 1$, and $\mu_0 = 0$.

This forms a birth and death process with parameters:

- $\lambda_0 = 3\lambda$, $\lambda_1 = 2\lambda$, $\lambda_2 = \lambda$, $\lambda_3 = 0$,
- $\mu_1 = \mu$, $\mu_2 = \mu$, $\mu_3 = \mu$.

However, the problem specifies distinct μ_i, so this doesn't apply generally.

Conclusion: Not a Birth and Death Process

Because the machines have different breakdown rates, the process cannot be modeled as a birth and death process when defining the state as the number of broken machines. The Markov property holds only if we track additional information, making the state space more complex.

Alternative Analysis: Continuous-Time Markov Chain

Instead, model the system as a continuous-time Markov chain with a state space capturing which machines are broken. Define the state as S, the set of broken machines, with state space as the power set of $\{1, 2, 3\}$ (8 states: $\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$):

- $S = \{\}$: All machines work. Transitions to $\{j\}$ at rate μ_j, for $j = 1, 2, 3$.
- $S \neq \{\}$, $k = |S|$:
  - **Breakdowns**: For each $j \in \{1, 2, 3\} \setminus S$, transition to $S \cup \{j\}$ at rate μ_j.
  - **Repairs**: The repairman works on one machine. Assuming a repair completion occurs at rate μ, and the fixed machine is chosen randomly (justifiable by the memoryless property and identical repair rates), transition to $S \setminus \{i\}$ for each $i \in S$ at rate μ/k.

Transition Rates Example

- $S = \{\}$: To $\{1\}$ at μ_1, $\{2\}$ at μ_2, $\{3\}$ at μ_3.
- $S = \{1\}$: To $\{1, 2\}$ at μ_2, $\{1, 3\}$ at μ_3, $\{\}$ at μ.

- $S = \{1, 2\}$: To $\{1, 2, 3\}$ at μ_3, $\{2\}$ at $\mu/2$, $\{1\}$ at $\mu/2$.
- $S = \{1, 2, 3\}$: To $\{2, 3\}$, $\{1, 3\}$, $\{1, 2\}$ each at $\mu/3$.

Analysis Approach

With this Markov chain:

1. **Transition Rate Matrix**: Construct the 8×8 generator matrix Q, where $q_{S,S'}$ is the transition rate from state S to S', and diagonal elements ensure row sums are zero.
2. **Steady-State Probabilities**: Solve $\pi Q = 0$ with $\sum \pi_S = 1$, where π_S is the long-run probability of state S.
3. **Performance Measures**: Compute metrics like the average number of broken machines or the repairman's utilization.

This approach captures the system's dynamics, accounting for distinct breakdown rates and the single repairman constraint.

Final Answer

We cannot analyze this system as a birth and death process because the transition rates depend on which specific machines are broken, not just their number, due to differing μ_i. Instead, we can analyze it as a continuous-time Markov chain with states as subsets of $\{1, 2, 3\}$, defining transition rates based on breakdown rates μ_i and repair rate μ, adjusted for the number of broken machines.

Q2

The machine can be in one of three states based on its operational status:

- **State U**: The machine is up and working.
- **State R1**: The machine is down and being repaired due to a type 1 failure.
- **State R2**: The machine is down and being repaired due to a type 2 failure.

The times spent in each state are exponentially distributed, and the transitions between states occur at specific rates:

- **From State U**: The machine is up for an exponentially distributed time with rate λ, meaning the mean up time is $1/\lambda$. Upon failure:
  - With probability p, it's a type 1 failure, and the machine moves to State R1. The transition rate from U to R1 is λp.
  - With probability $1 - p$, it's a type 2 failure, and the machine moves to State R2. The transition rate from U to R2 is $\lambda(1 - p)$.
- **From State R1**: The repair time for a type 1 failure is exponential with rate μ_1 (mean repair time $1/\mu_1$), after which the machine returns to State U. The transition rate from R1 to U is μ_1.
- **From State R2**: The repair time for a type 2 failure is exponential with rate μ_2 (mean repair time $1/\mu_2$), after which the machine returns to State U. The transition rate from R2 to U is μ_2.

Since this is a continuous-time Markov chain with exponential holding times, the long-run proportion of time spent in each state corresponds to the stationary probabilities, denoted:

- π_U: Proportion of time in State U (up).
- π_{R1}: Proportion of time in State R1 (down due to type 1 failure).
- π_{R2}: Proportion of time in State R2 (down due to type 2 failure).

Balance Equations

- **State U**:
  - Rate out: $\pi_U \cdot (\lambda p + \lambda(1 - p)) = \pi_U \cdot \lambda$.
  - Rate in: $\pi_{R1} \cdot \mu_1 + \pi_{R2} \cdot \mu_2$ (entering from R1 and R2).
  - Equation: $\pi_U \lambda = \pi_{R1} \mu_1 + \pi_{R2} \mu_2$.
- **State R1**:
  - Rate out: $\pi_{R1} \cdot \mu_1$.
  - Rate in: $\pi_U \cdot \lambda p$.
  - Equation: $\pi_{R1} \mu_1 = \pi_U \lambda p$.
- **State R2**:
  - Rate out: $\pi_{R2} \cdot \mu_2$.
  - Rate in: $\pi_U \cdot \lambda(1 - p)$.
  - Equation: $\pi_{R2} \mu_2 = \pi_U \lambda(1 - p)$.

$$\pi_U \left(1 + \frac{\lambda p}{\mu_1} + \frac{\lambda(1 - p)}{\mu_2}\right) = 1$$

Solve for π_U, π_{R1} and π_{R2}:

$$\pi_U = \frac{1}{1 + \frac{\lambda p}{\mu_1} + \frac{\lambda(1-p)}{\mu_2}}$$

$$\pi_{R1} = \frac{\pi_U \lambda p}{\mu_1} = \frac{\frac{\lambda p}{\mu_1}}{1 + \frac{\lambda p}{\mu_1} + \frac{\lambda(1-p)}{\mu_2}}$$

$$\pi_{R2} = \frac{\pi_U \lambda (1-p)}{\mu_2} = \frac{\frac{\lambda(1-p)}{\mu_2}}{1 + \frac{\lambda p}{\mu_1} + \frac{\lambda(1-p)}{\mu_2}}$$

Q3

Since arrivals follow a Poisson process and service times are exponential, the system's state, defined as the number of customers present (denoted $N(t)$), evolves as a continuous-time Markov chain. The possible states are $0, 1, 2, \ldots$, representing the total number of customers in the system. Transitions occur between adjacent states:

- **Upward transitions** ($n \to n+1$): A new customer arrives and joins.
- **Downward transitions** ($n \to n-1$, for $n \geq 1$): The server completes a service.

Define the Transition Rates

- **Arrival Rate (Birth Rate):** The rate at which customers join when the system is in state n is:

$$\lambda_n = \lambda \cdot \frac{1}{n+1}$$

  This is the transition rate from n to $n+1$.
- **Service Rate (Death Rate):** The service rate is μ whenever there's at least one customer ($n \geq 1$), and 0 when the system is empty ($n = 0$). So:

$$\mu_n = \mu \quad \text{for } n \geq 1$$

  This is the transition rate from n to $n-1$.

The Markov chain is a birth-death process because transitions are only between consecutive states:

- From n to $n+1$ at rate $\lambda_n = \lambda/(n+1)$,
- From n to $n-1$ at rate $\mu_n = \mu$ (for $n \geq 1$),
- No transitions from 0 to -1, so $\mu_0 = 0$.

Detailed balance equates the flow from n to $n+1$ with the flow from $n+1$ to n:

$$\lambda_n \pi_n = \mu_{n+1} \pi_{n+1} \quad \text{for } n = 0, 1, 2, \ldots$$

Substituting our rates:

$$\frac{\lambda}{n+1} \pi_n = \mu \pi_{n+1}$$

$$\therefore \pi_{n+1} = \frac{\lambda}{n+1} \cdot \frac{1}{\mu} \pi_n = \frac{\lambda}{\mu} \cdot \frac{1}{n+1} \pi_n$$

Starte with π_0:

- For $n = 0$:

$$\pi_1 = \frac{\lambda}{\mu} \cdot \frac{1}{1} \pi_0 = \frac{\lambda}{\mu} \pi_0$$

- For $n = 1$:

$$\pi_2 = \frac{\lambda}{\mu} \cdot \frac{1}{2} \pi_1 = \frac{\lambda}{\mu} \cdot \frac{1}{2} \cdot \frac{\lambda}{\mu} \pi_0 = \left(\frac{\lambda}{\mu}\right)^2 \cdot \frac{1}{2} \pi_0$$

- For $n = 2$:

$$\pi_3 = \frac{\lambda}{\mu} \cdot \frac{1}{3} \pi_2 = \frac{\lambda}{\mu} \cdot \frac{1}{3} \cdot \left(\frac{\lambda}{\mu}\right)^2 \cdot \frac{1}{2} \pi_0 = \left(\frac{\lambda}{\mu}\right)^3 \cdot \frac{1}{3 \cdot 2} \pi_0$$

Generally:

$$\pi_n = \pi_0 \cdot \left(\frac{\lambda}{\mu}\right)^n \cdot \frac{1}{n!} \cdot (n-1) \cdot \ldots \cdot 1$$

The product $n \cdot (n-1) \cdot \ldots \cdot 1 = n!$, so:

$$\pi_n = \pi_0 \cdot \frac{\left(\frac{\lambda}{\mu}\right)^n}{n!}$$

∵ The probabilities must sum to 1:

$$\sum_{n=0}^{\infty} \pi_0 \cdot \frac{\left(\frac{\lambda}{\mu}\right)^n}{n!} = \pi_0 \sum_{n=0}^{\infty} \frac{\left(\frac{\lambda}{\mu}\right)^n}{n!} = \pi_0 \cdot e^{\lambda/\mu} = 1$$

$$\therefore \pi_0 = e^{-\lambda/\mu}$$

$$\therefore \pi_n = e^{-\lambda/\mu} \cdot \frac{\left(\frac{\lambda}{\mu}\right)^n}{n!}$$

This is the pmf of a Poisson distribution with mean λ/μ.

Q4

Define the state of the system as n, the number of machines not in use (i.e., either being repaired or waiting). Since there are three machines, $n = 0, 1, 2, 3$:

- $n = 0$: All three machines are working.
- $n = 1$: One machine is being repaired by one repairman; two are working.
- $n = 2$: Two machines are being repaired by two repairmen; one is working.
- $n = 3$: Two machines are being repaired by two repairmen; one is waiting.

This is a birth-death process:

- **Breakdown (birth) rate**: When n machines are down, $3 - n$ are working. Each working machine breaks down at rate $\lambda = 1/10 = 0.1$, so the total breakdown rate from state n to $n + 1$ is:

$$\lambda_n = (3 - n)\lambda \quad \text{for} \quad n = 0, 1, 2$$

  - $\lambda_0 = 3\lambda = 0.3$
  - $\lambda_1 = 2\lambda = 0.2$
  - $\lambda_2 = \lambda = 0.1$
  - $\lambda_3 = 0$ (no working machines to break down)
- **Repair (death) rate**: Each repairman fixes a machine at rate $\mu = 1/8 = 0.125$ (mean time is 8). The repair rate depends on the number of busy repairmen:
  - $\mu_0 = 0$ (no repairs when all machines are working)
  - $\mu_1 = \mu = 0.125$ (one repairman busy)
  - $\mu_2 = 2\mu = 0.25$ (two repairmen busy)
  - $\mu_3 = 2\mu = 0.25$ (two repairmen busy, one machine waiting)

Steady-State Probabilities

For a birth-death process with states $n = 0, 1, \ldots, N$ (here $N = 3$), the steady-state probability π_n is:

$$\pi_n = \pi_0 \prod_{k=1}^{n} \frac{\lambda_{k-1}}{\mu_k}, \quad \sum_{n=0}^{N} \pi_n = 1$$

Compute the ratios with $\lambda = 0.1$, $\mu = 0.125$:

- $\pi_1 = \pi_0 \frac{\lambda_0}{\mu_1} = \pi_0 \cdot \frac{12}{5}$
- $\pi_2 = \pi_1 \frac{\lambda_1}{\mu_2} = \pi_0 \cdot \frac{48}{25}$
- $\pi_3 = \pi_2 \frac{\lambda_2}{\mu_3} = \pi_0 \cdot \frac{96}{125}$

Normalization:

$$\pi_0 \left[1 + \frac{12}{5} + \frac{48}{25} + \frac{96}{125}\right] = 1$$

$$\therefore \pi_0 = \frac{125}{761}$$

Then:

- $\pi_1 = \frac{1500}{761}$
- $\pi_2 = \frac{240}{761}$
- $\pi_3 = \frac{96}{761}$

(a) Average Number of Machines Not in Use

The average number of machines not in use is the expected value of n:

$$E[n] = 0 \cdot \pi_0 + 1 \cdot \pi_1 + 2 \cdot \pi_2 + 3 \cdot \pi_3 = \frac{1068}{761}$$

(b) Proportion of Time Both Repairmen Are Busy

Both repairmen are busy when $n \geq 2$, i.e., $n = 2$ or 3:

$$\pi_2 + \pi_3 = \frac{336}{761}$$

Q5

(a) Define states so as to analyze the preceding as a continuous-time Markov chain

Let:

- **State space** S: The power set of $\{1, 2, \ldots, n\}$, i.e., all possible subsets of the server indices, including the empty set $\emptyset$. A state $A \subseteq \{1, 2, \ldots, n\}$ indicates that the servers in A are busy, and those in $\{1, 2, \ldots, n\} \setminus A$ are idle.

Transitions occur due to:

1. **Arrivals**: If $|A| < n$ (at least one idle server), a customer arrives at rate λ and chooses one of the $n - |A|$ idle servers uniformly. For each $i \notin A$, the transition rate from A to $A \cup \{i\}$ is:
$$q(A, A \cup \{i\}) = \frac{\lambda}{n - |A|}$$
If $|A| = n$, arrivals leave, so no state change occurs due to arrivals.
2. **Departures**: For each busy server $i \in A$, service completes at rate μ_i, moving the state from A to $A \setminus \{i\}$. The transition rate is:
$$q(A, A \setminus \{i\}) = \mu_i \quad \text{for} \quad i \in A$$

This defines a CTMC with state space $S = 2^{\{1,2,\ldots,n\}}$, where transitions reflect the system's dynamics based on arrivals and departures.

(b) Show that this chain is time reversible

The possible transitions are:

- **Arrival**: $A \to A \cup \{i\}$ for $i \notin A$, $|A| < n$, with rate $q(A, A \cup \{i\}) = \frac{\lambda}{n-|A|}$.
- **Departure**: $A \to A \setminus \{i\}$ for $i \in A$, with rate $q(A, A \setminus \{i\}) = \mu_i$.

Check Detailed Balance

1. **Arrival Transition**: Consider A with $|A| < n$ and $i \notin A$, so $B = A \cup \{i\}$.
   - Forward: $q(A, A \cup \{i\}) = \frac{\lambda}{n-|A|}$.
   - Reverse: From $A \cup \{i\}$ to A, server i completes service, so $q(A \cup \{i\}, A) = \mu_i$.
   - Detailed balance: $\pi_A \cdot \frac{\lambda}{n-|A|} = \pi_{A\cup\{i\}} \cdot \mu_i$.
2. **Departure Transition**: Consider A with $i \in A$, so $B = A \setminus \{i\}$.
   - Forward: $q(A, A \setminus \{i\}) = \mu_i$.
   - Reverse: From $A \setminus \{i\}$ to A, an arrival occupies server i (since $i \notin A \setminus \{i\}$), with $|A \setminus \{i\}| = |A| - 1$, so $q(A \setminus \{i\}, A) = \frac{\lambda}{n-(|A|-1)}$.
   - Detailed balance: $\pi_A \cdot \mu_i = \pi_{A\setminus\{i\}} \cdot \frac{\lambda}{n-(|A|-1)}$.

Suppose:

$$\pi_{A\cup\{i\}} = \pi_A \cdot \frac{\lambda/(n - |A|)}{\mu_i}$$

This suggests a recursive form. Starting from $\pi_\emptyset$, for $A = \{i_1, i_2, \ldots, i_m\}$ built by adding servers, at step k, the number of busy servers is $k - 1$, so:

$$\pi_A = \pi_\emptyset \prod_{k=1}^{m} \frac{\lambda/(n - (k-1))}{\mu_{i_k}} = \pi_\emptyset \cdot \frac{\lambda^m}{\prod_{k=1}^{m}(n-k+1) \cdot \prod_{i\in A} \mu_i}$$

Verify both equations with this π_A:

- **Arrival**: $\pi_{A\cup\{i\}} = \pi_\emptyset \cdot \frac{\lambda^{|A|+1}}{\prod_{k=1}^{|A|+1}(n-k+1)\cdot\prod_{j\in A\cup\{i\}}\mu_j}$, and $\pi_A = \pi_\emptyset \cdot \frac{\lambda^{|A|}}{\prod_{k=1}^{|A|}(n-k+1)\cdot\prod_{j\in A}\mu_j}$, so:
$$\pi_A \cdot \frac{\lambda}{n - |A|} = \pi_{A\cup\{i\}} \cdot \mu_i$$
holds, as the terms match when substituting.
- **Departure**: Similarly, it holds (as derived consistently in the thinking trace).

Since the state space is finite and irreducible ($\lambda > 0$, $\mu_i > 0$), and detailed balance is satisfied for all transition pairs, the chain is time reversible.

(c) Find the limiting probabilities

For a finite, irreducible CTMC, the limiting probabilities equal the stationary distribution. From (b):

$$\pi_A = \pi_\emptyset \cdot \frac{\lambda^{|A|}}{\prod_{k=1}^{|A|}(n-k+1) \cdot \prod_{i \in A} \mu_i}$$

where $\prod_{k=1}^{|A|}(n-k+1) = n(n-1)\cdots(n-|A|+1)$, and $\pi_\emptyset$ is the normalizing constant. Since $\sum_{A \subseteq \{1,2,\ldots,n\}} \pi_A = 1$:

$$\pi_\emptyset \sum_{A} \frac{\lambda^{|A|}}{\prod_{k=1}^{|A|}(n-k+1) \cdot \prod_{i \in A} \mu_i} = 1$$

Thus:

$$\pi_\emptyset = \left[\sum_{A \subseteq \{1,2,\ldots,n\}} \frac{\lambda^{|A|}}{\prod_{k=1}^{|A|}(n-k+1) \cdot \prod_{i \in A} \mu_i} \right]^{-1}$$

For $A = \emptyset$, $|A| = 0$, the numerator is $\lambda^0 = 1$, and the denominator's product over k is 1 (empty product), with $\prod_{i \in A} \mu_i = 1$, so $\pi_\emptyset$ is consistent.

Final Limiting Probabilities:

$$\pi_A = \frac{\frac{\lambda^{|A|}}{\prod_{k=1}^{|A|}(n-k+1) \cdot \prod_{i \in A} \mu_i}}{\sum_{B \subseteq \{1,2,\ldots,n\}} \frac{\lambda^{|B|}}{\prod_{k=1}^{|B|}(n-k+1) \cdot \prod_{i \in B} \mu_i}}$$

Q6

Part (a): Probability of Exactly 4 Customers in the System

- λ = 5 per minute (arrival rate),
- μ = 3 per minute (service rate, since 1/μ = 1/3 minutes implies μ = 1 / (1/3) = 3),
- ρ = λ/μ = 5/3.

For a Poisson random variable N with parameter ρ, the probability of exactly n customers is:

$$P(N = n) = \frac{\rho^n}{n!} e^{-\rho}$$

For $n = 4$ and ρ = 5/3:

$$P(N = 4) = \frac{\left(\frac{5}{3}\right)^4}{4!} e^{-\frac{5}{3}} = \frac{625}{1944} e^{-\frac{5}{3}}$$

Part (b): Expected Number of Customers in the System

$$E[N] = \lambda/\mu = 5/3$$

Part (c): Variance of the Number of Customers in the System

$$\mathrm{Var}(N) = \rho = \lambda/\mu = 5/3$$